FROB C5 full quotient Frobenius maps

Validation audit, 22 October 2026

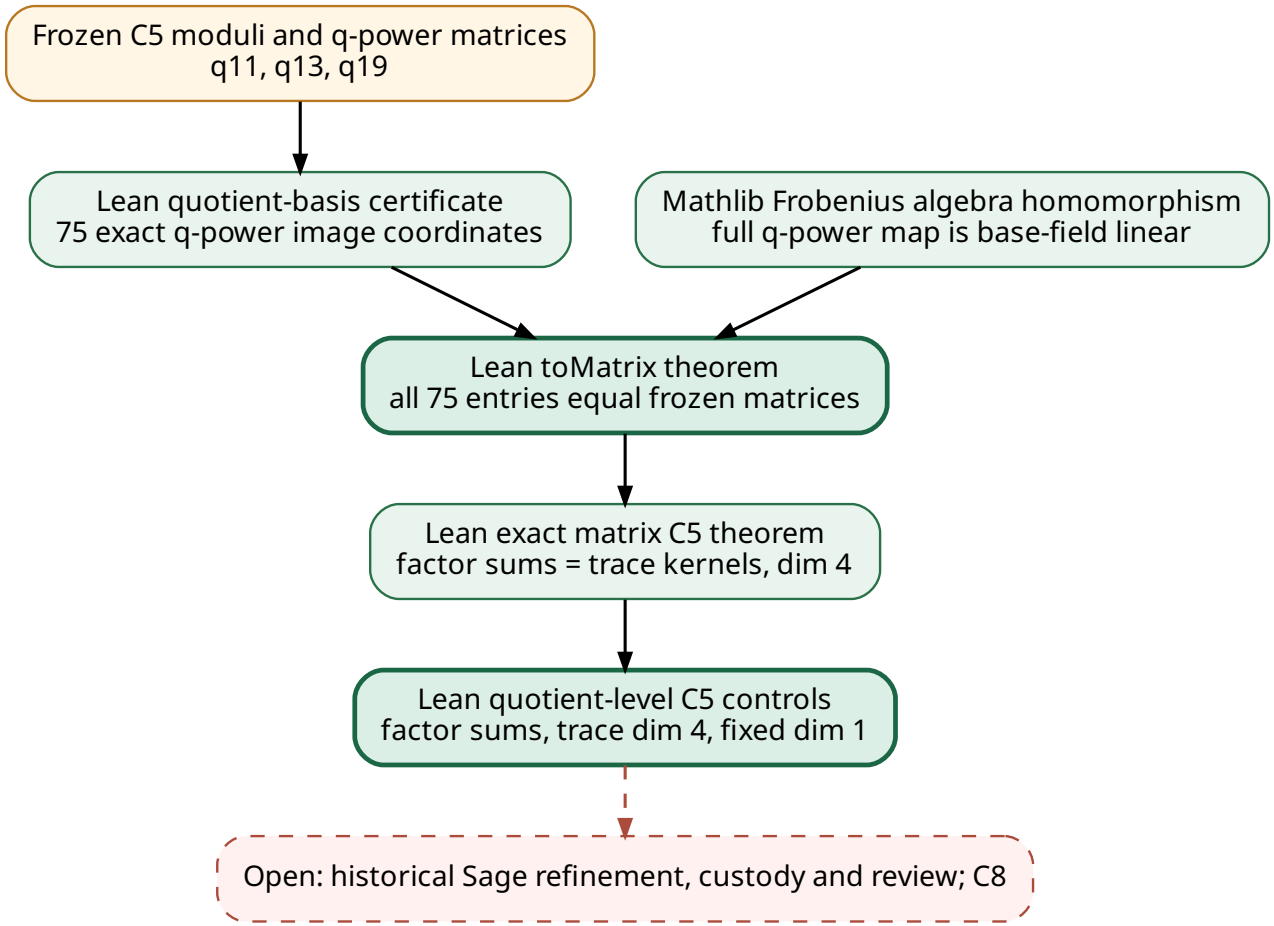
Scope. For each frozen polynomial quotient over $\mathbf{F}_{11}$, $\mathbf{F}_{13}$, or $\mathbf{F}_{19}$, Lean identifies the matrix of the full q-power endomorphism in its power basis. This closes the gap between previously proved basis-image coordinates and a linear-map matrix.

Base field	Frozen quotient modulus	Matrix equality
$\mathbf{F}_{11}$	$z^5 + 2z^4 + 1$	all 5×5 entries
$\mathbf{F}_{13}$	$z^5 + 8z^4 + 1$	all 5×5 entries
$\mathbf{F}_{19}$	$z^5 + 3z^4 + 1$	all 5×5 entries

Proof. Mathlib's `frobeniusAlgHom` raises each quotient ring element to its base-field cardinality and supplies an algebra homomorphism, hence a base-field-linear map. Lean proves the quotient moduli have degree five, uses the `AdjoinRoot` power basis, and rewrites each `toMatrix` entry as the coordinate of the q-power image of a basis monomial. The earlier exact quotient certificate proves that coordinate equals the frozen matrix entry. All 75 full-map entries are conjoined in one target theorem. No irreducibility assumption is needed for this ring-level matrix calculation.

Machine verification. Source commit `1993b13e4a24` uses pinned Lean 4.35.0-rc1. The exact-source verifier passed an 8,975-job full build and 3,093-job target build. Its global audit found 667 theorem declarations with standard axioms only; the requested theorem audit passed. Independent semantic review remains pending.

Boundary. This is a mathematical quotient-algebra map result. An additional Lean certificate proves the corresponding C5 controls for the actual quotient q-power endomorphisms. Historical Sage field and kernel paths are not proved to refine it; original execution-source custody and independent code review remain open. The three moduli have not been proved irreducible in Lean, although the independent source-bound replay checks them. C8, the complete experiment gate, and performance claims remain open. No trial or timing run was repeated.


Frozen C5 moduli and q-power matrices
q11, q13, q19
Lean quotient-basis certificate
75 exact q-power image coordinates
Mathlib Frobenius algebra homomorphism
full q-power map is base-field linear
Lean toMatrix theorem
all 75 entries equal frozen matrices
Lean exact matrix C5 theorem
factor sums = trace kernels, dim 4
Lean quotient-level C5 controls
factor sums, trace dim 4, fixed dim 1
Open: historical Sage refinement, custody and review; C8